MOBILE COMPUTING

COMPLETE EXAMINATION STUDY GUIDE

College of Business Education | ITU07401

Questions & Model Answers | Past Paper Solutions | Tutorial Format

Topics Covered
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📝 *How to Use: Every Q&A follows the WINNING formula: Define → Explain → Real Example → Conclude. CBE past paper questions (green boxes) appear exactly as set in the exam. Follow the exam strategy reminder on the last page.*

UNIT 1: MAC Protocols — Wireless Fundamentals

1.1 Introduction & Key Terminology

Term	Definition
MAC	Medium Access Control — sublayer controlling how devices gain access to the shared wireless channel.
Collision	Two+ nodes transmit simultaneously → signals overlap → unreadable at receiver.
CSMA/CA	Carrier Sense Multiple Access/Collision Avoidance — listen before transmitting, avoid collisions.
TDMA	Time Division Multiple Access — each node gets a fixed time slot.
RTS/CTS	Request to Send / Clear to Send — handshake to reserve the channel before data.
Hidden Node	In range of receiver but OUT of range of sender → causes surprise collisions at receiver.
Exposed Node	In range of sender but OUT of range of destination → needlessly defers from transmitting.
NAV	Network Allocation Vector — timer set by CTS/RTS, telling a node to stay silent.

Q1: What is the MAC protocol and why is it needed in wireless networks?

✅ **Answer**

MAC (Medium Access Control) defines rules by which multiple nodes share a single wireless channel. It is needed because:

- The wireless channel is shared — simultaneous transmissions collide and are lost.
- Wireless nodes CANNOT detect collisions while transmitting (no CSMA/CD — half-duplex).
- Without MAC: constant collisions, wasted bandwidth, idle listening, and interference.
- The wireless channel is time-varying (fading), making access control more complex.

📝 **Exam Tip:** In exams always mention half-duplex, hidden node, and time-varying channel as reasons why wireless MAC is harder than wired.

Q2: Explain the THREE major wireless MAC challenges.

✅ **Answer**

1. Half-Duplex Operation:

A node cannot transmit and receive simultaneously. The transmitter's powerful signal leaks into its own receiver, drowning incoming signals. This makes collision detection (CSMA/CD used in Ethernet) impossible. All wireless MAC focuses on collision AVOIDANCE instead.

2. Time-Varying Channel:

Due to reflection, diffraction, and scattering, the received signal is a sum of time-shifted copies (multipath propagation). This causes fading — a node in fade may not hear the channel and wrongly assume it is free.

3. Burst Channel Errors:

Wireless BER can reach 10^-3 vs. wired 10^-6. Errors occur in bursts during fades. Solutions: smaller packets, Forward Error Correction (FEC), retransmissions (ACKs).

Q3: Explain the Hidden Terminal and Exposed Terminal Problems with examples.

✅ **Answer**

Hidden Terminal Problem:

Setup: A ———► B ◄——— C (A and C cannot hear each other; both can reach B)

- A transmits to B.
- C cannot detect A's carrier → C falsely thinks channel is idle → C also transmits.
- COLLISION at B. Neither A nor C knows. Throughput and reliability are reduced.
- Solution: RTS/CTS. A sends RTS → B replies with CTS heard by C → C stays silent.

Exposed Terminal Problem:

Setup: A ◄——— B ———► C ———► D (D is out of B's range)

- B is transmitting to A. C wants to transmit to D.
- C detects B's signal → C assumes channel is busy → C waits unnecessarily.
- C's transmission to D would NOT interfere at A (D is out of B's range).
- Result: Wasted bandwidth — C is exposed to B but could safely transmit.
- KEY DIFFERENCE: Hidden terminal = under-deferral (collision occurs). Exposed terminal = over-deferral (no collision, just wasted bandwidth).

📝 **Exam Tip:** Draw the A-B-C diagrams in exams! State clearly: Hidden = collision risk; Exposed = wasted opportunity.

Q4: How are RTS and CTS used to avoid collisions? (CBE-style 6 marks)

✅ **Answer**

RTS (Request to Send) and CTS (Clear to Send) are signalling packets in IEEE 802.11 used to RESERVE the wireless channel BEFORE sending actual data.

Step-by-Step Process:

- Step 1 — RTS: Sender transmits short RTS to receiver containing: sender address, receiver address, and channel reservation DURATION.
- Step 2 — CTS: Receiver replies with CTS — also broadcast to ALL nearby devices, containing the duration field.
- Step 3 — NAV: ALL devices hearing CTS set their NAV timer and stay silent for the duration — even hidden terminals that could not hear the RTS.
- Step 4 — Data Transmission: Sender transmits the data packet, confident no other device will interfere.
- Step 5 — ACK: Receiver sends ACK. Channel is released.

Why it prevents collisions: A hidden terminal C (cannot hear A's RTS) DOES hear B's CTS → sets NAV → stays silent. If RTS or CTS collide, only tiny control frames are wasted — not large data packets.

UNIT 2: WSN MAC Protocols — Energy Efficiency

2.1 Energy Waste Sources

Waste Source	Description	Impact
Idle Listening	Radio kept ON even when no data expected. Nearly same energy as active receive.	Major energy waster in WSN
Collisions	Two signals arrive simultaneously → both lost. Transmission energy wasted.	Especially costly in multi-hop nets
Overhearing	Node receives packets NOT destined for it.	Wastes processing and energy
Protocol Overhead	RTS/CTS frames carry no data but cost energy. Can be 70% of total traffic.	High in dense networks
Overmitting	Data sent to a node not ready to receive it.	Wastes energy on useless transmissions
Traffic Fluctuation	Fixed duty cycles mismatch variable traffic.	Static protocols waste energy in idle

Q5: What is S-MAC? Explain its key components and limitations.

✅ **Answer**

S-MAC (Sensor-MAC) is an energy-efficient MAC for WSNs, inspired by IEEE 802.11, using periodic listen/sleep to reduce idle listening.

Four Key Components:

- Periodic Listen & Sleep: Nodes alternate between fixed 115ms active periods and variable sleep. Duty cycle = determined by sleep length.
- Time Synchronization: SYNC packets exchanged at start of each active period so neighbors share sleep schedules.
- Collision Avoidance: RTS/CTS mechanism to handle hidden terminals.
- Overhearing Avoidance: When a node hears RTS/CTS not addressed to it, it sleeps for the expected transmission duration.

Limitations:

- Poor scalability — as network grows, S-MAC must maintain increasing numbers of neighbor schedules.
- Fairness issue — a node with lots of data can monopolize the channel.
- Does NOT adapt to traffic load — fixed sleep intervals waste energy when traffic is low.

📝 **Exam Tip:** S-MAC is the most examined WSN protocol. Always list all 4 components.

Contrast with T-MAC for variable traffic questions.

Q6: Compare Contention-Based vs. TDMA-Based MAC protocols for WSNs.

✅ **Answer**

Feature	Contention-Based (S-MAC, B-MAC)	TDMA-Based (LMAC, A-MAC)
Collisions	Possible — CSMA/CA reduces them	Zero — each node has its own slot
Idle listening	Reduced via sleep schedules	Eliminated — sleep outside own slot
Synchronization	Lightweight	Strict — costly in hardware
Scalability	Better — decentralized	Poor — frame length hard to change
Latency	Low to medium	Increases linearly with nodes
Best for	Variable, decentralized networks	Energy-constrained, delay-sensitive

UNIT 3: The OSI Model

3.1 What is the OSI Model?

OSI (Open Systems Interconnection) is a 7-layer framework created by ISO in 1984 to standardize how data travels between computers on any network. Each layer has ONE specific job and communicates only with the layers directly above and below it.

Memory Trick (top→bottom): 'All People Seem To Need Data Processing' — Application, Presentation, Session, Transport, Network, Data Link, Physical.

Layer #	Layer Name	Job	Key Protocols	Simple Trick
7	Application	Network services to user apps	HTTP, HTTPS, FTP, DNS, SMTP	The app you USE
6	Presentation	Translates, encrypts, compresses	TLS/SSL, JPEG, MPEG	The TRANSLATOR
5	Session	Opens/manages/ closes sessions	NetBIOS, RPC	The SESSION manager
4	Transport	Reliable end-to-end delivery, segments	TCP, UDP	The DELIVERY guarantee
3	Network	Routes packets across networks using IP	IP, ICMP, OSPF	The NAVIGATOR
2	Data Link	Node-to-node using MAC addresses	Ethernet, WiFi 802.11	The LOCAL postman
1	Physical	Transmits raw bits over physical medium	Cables, radio waves	The WIRE/SIGNAL

Q7: Explain the role of the Transport Layer in the OSI model.

✅ **Answer — 4-step model answer**

Step 1 — DEFINE: The Transport Layer (Layer 4) is responsible for reliable end-to-end delivery of data between source and destination devices across the network.

Step 2 — EXPLAIN HOW: It breaks large data into segments, assigns sequence numbers for correct reassembly, uses port numbers (e.g., HTTPS=443) to direct data to the correct application, and uses TCP for guaranteed delivery or UDP for speed without guarantee.

Step 3 — REAL EXAMPLE: When a student downloads a university assignment PDF from Moodle, TCP at Layer 4 breaks the file into hundreds of segments. If any segment is lost en route, TCP automatically retransmits it. The student receives a complete, uncorrupted PDF.

Step 4 — CONCLUDE: The Transport Layer is essential because it ensures data sent over an unreliable network arrives correctly and completely, regardless of what happens in the network infrastructure below.

Q8: Compare TCP and UDP.

✅ **Answer**

Feature	TCP	UDP
Connection	3-way handshake required first	No connection — send immediately
Reliability	Guaranteed — retransmits lost packets	Not guaranteed — lost = gone
Speed	Slower (due to ACKs)	Faster — no waiting
Order	Correct order guaranteed	May arrive out of order
Overhead	Higher — headers + ACKs	Lower — minimal headers
Use cases	File download, banking, email, web	Live video, voice calls, gaming, DNS
Real example	Downloading PDF from Moodle	Watching live football on YouTube

Q9: Compare the OSI model and the TCP/IP model.

✅ **Answer**

OSI Layer	TCP/IP Layer	Protocols
Application (7)	Application	HTTP, HTTPS, FTP, DNS, SMTP, MQTT
Presentation (6)	Application	SSL/TLS, JPEG, MPEG
Session (5)	Application	NetBIOS, RPC, SIP
Transport (4)	Transport	TCP, UDP
Network (3)	Internet	IP, ICMP, OSPF, BGP
Data Link (2)	Network Interface	Ethernet, WiFi 802.11
Physical (1)	Network Interface	Copper, Fibre, Radio waves

Key distinction: OSI = 7-layer THEORETICAL model for learning. TCP/IP = 4-layer PRACTICAL model that the internet actually uses. OSI helps you understand — TCP/IP is what networks actually run.

UNIT 4: Introduction to Mobile Computing

Q10: Define the term Mobile Computing.

✅ **Answer**

Mobile Computing is a technology that allows transmission of data, voice, and video via a computer or any other wireless-enabled device without having to be connected to a fixed physical link.

In simple terms: it is the ability to use computing devices (phones, tablets, laptops) while physically moving, while remaining connected to a network and accessing or exchanging information in real time.

Real Example: A doctor in a moving ambulance accessing a patient's medical records on a tablet — the doctor is moving, the device is wireless, and live data is being accessed. That is mobile computing.

Q11: Explain FOUR characteristics and TWO constraints of mobile computing.

✅ **Answer**

FOUR CHARACTERISTICS:

- Mobility — Users can physically move from one location to another while remaining connected. The network service continues without interruption. Example: A sales executive checking emails while commuting by train.
- Portability — Devices are small and lightweight enough to be carried everywhere — smartphones, tablets, laptops. Example: A field engineer carrying a tablet to access engineering drawings remotely.
- Wireless Communication — Uses WiFi, 3G/4G/5G, Bluetooth. No physical cables connect the device. Example: Browsing the internet on a phone using 4G coverage.
- Connectivity — Device automatically switches between available networks (WiFi to cellular) to maintain connectivity. Example: A smartphone switching from home WiFi to 4G as you leave home without interrupting a video stream.

TWO CONSTRAINTS:

- Limited Resources — Mobile devices have less processing power, memory, storage, and battery than desktops. Batteries must be recharged regularly. Example: A smartphone running a GPS mapping app may last only 3–4 hours on battery.
- Variable Connectivity — Signal strength changes constantly as the user moves. Network performance varies with location, obstacles, interference, and number of users sharing the network. Example: A video call clear at home may break up in an area with weak 4G signal.

Q12: Differentiate between User Mobility and Host Mobility.

✅ **Answer**

Aspect	User Mobility	Host Mobility
Definition	Ability of a PERSON to move and continue using the same service with the same identity/account.	Ability of a DEVICE (host) to move between networks while maintaining a consistent reachable network address.
What moves?	The person — device may or may not change.	The device itself moves between networks.
IP handling	User identity follows via login — IP may change without problem.	Must maintain reachable IP using Mobile IP (Home Agent, COA, tunneling).
Complexity	Less complex — service follows login credentials.	More complex — requires Mobile IP to keep network identity stable.
Example	Student moves from Dar es Salaam to Mwanza and accesses university portal with same credentials.	A server on a vehicle that moves between cities but remains reachable at its permanent IP.

Q13: Differentiate between FDMA and TDMA.

✅ **Answer**

Feature	FDMA	TDMA
Full name	Frequency Division Multiple Access	Time Division Multiple Access
How it works	Divides BANDWIDTH into separate frequency channels. Each user gets own dedicated frequency.	Divides TIME into slots. All users share one frequency but take turns per slot.
Simultaneous use	All users transmit simultaneously — on different frequencies.	Users take turns — only one per time slot.
Guard mechanism	Guard BANDS between adjacent frequencies.	Guard TIMES between adjacent slots.
Synchronization	Not required between users.	Strict timing synchronization required.
Power efficiency	Lower — dedicated channel wasted when user is idle.	Higher — unused slots can be reallocated.
Primary use	Early cellular; GSM uses FDMA for gross frequency division.	GSM (2G), PDC, IS-136 mobile systems.
Analogy	Multiple radio stations — each on own frequency permanently.	Meeting: each person speaks for 30 seconds, same microphone, in turns.

GSM fact: GSM combines BOTH — FDMA divides spectrum into 200 kHz channels, then TDMA divides each channel into 8 time slots (8 users per frequency).

Q14: Differentiate between Mobility and Nomadicity.

✅ **Answer**

Aspect	Mobility	Nomadicity
Definition	Capability to physically move or switch between networks while maintaining connectivity.	Computing paradigm where the entire environment (apps, data, preferences) travels WITH the user — accessible from any location or device.
Focus	The ACT of movement.	CONTINUITY of computing experience.
Technical need	Wireless networks, Mobile IP, handover mechanisms.	Cloud storage, VPNs, sync tools, device-independence.
Example	Walking while on a 4G video call — towers switch automatically.	Professor carrying a laptop with all notes and grades everywhere.

UNIT 5: Mobile IP

5.1 Overview

Mobile IP allows a mobile device to keep the SAME IP address and maintain active connections as it moves between networks. The internet was designed for stationary devices — IP addresses are tied to network locations. Mobile IP solves this.

Without Mobile IP: Moving to a new network breaks ALL active connections — calls drop, downloads fail, streaming stops.

With Mobile IP: The Home Agent transparently forwards traffic to the new network — connections continue seamlessly.

Term	Definition
MN (Mobile Node)	Your device — has a permanent home IP that never changes.
HA (Home Agent)	Router in your home network. Intercepts incoming traffic for MN and tunnels it to the current location.
FA (Foreign Agent)	Router in the visited network. Receives tunneled packets from HA and delivers to MN.
COA (Care-of Address)	Temporary IP address representing the MN's current location in the foreign network.
Tunneling	HA wraps the original packet in a new packet addressed to the COA — 'IP-in-IP'.
Binding	MN's registration of its current COA with the HA so HA can tunnel traffic correctly.

Q15: What is a Care-of Address (COA)? Explain its two types.

✅ **Answer**

A Care-of Address (COA) is a temporary IP address assigned to a Mobile Node when visiting a foreign network. It represents the MN's CURRENT LOCATION. The Home Agent uses the COA to tunnel data to the MN wherever it is.

Key point: COA is never stored in DNS — other nodes always look up the MN's permanent home address. The HA handles forwarding transparently.

Type	Description	How Obtained	Key Feature
Foreign Agent COA (FA-COA)	The IP address of the Foreign Agent router IS the COA. FA receives all tunneled packets from HA and delivers to visiting MN.	From FA's periodic Agent Advertisement messages.	SHARED — multiple MNs visiting same network share one FA-COA. Saves IP addresses.

Co-located COA (CCoA)	A unique temporary IP assigned DIRECTLY to the MN's own interface — not the FA.	Via DHCP from foreign network.	UNIQUE per MN. Allows route optimization. Works even with no FA in foreign network.

Q16: Explain triangular routing problems and the solution.

✅ **Answer**

Triangular Routing is the default incoming data path in Mobile IP:

Correspondent Node (CN) → Home Agent (HA) → Foreign Agent (FA) → Mobile Node (MN)

Called 'triangular' because data goes to the HA first — even if CN and MN are geographically close.

Example: Kenyan student (MN) studying in London. Friend (CN) in Nairobi sends data. Path: Nairobi (CN) → Nairobi (HA) → London (FA) → Student. The message travels to HA and back when sender and HA are both in Nairobi.

Problems with Triangular Routing:

- Sub-optimal routing — data takes a long, inefficient path even when CN and MN are nearby.
- Increased latency — extra hops through HA add delay. Terrible for voice/video calls.
- HA bottleneck — all roaming traffic passes through HA → single point of congestion and failure.
- Firewall problem (Ingress Filtering) — MN's outgoing data has a home IP source address; foreign network firewall may reject it.
- Multicast problem — MN loses group membership if it sends directly from foreign network.
- TTL problem — packet TTL may expire before completing the longer triangular path.

Solution: Reverse Tunneling

Creates a reverse tunnel MN → FA → [tunnel] → HA → CN for OUTGOING traffic. This fixes ingress filtering, multicast, and TTL problems.

Better Solution: Route Optimization (RFC 3775) — MN sends Binding Update directly to CN. CN stores COA in its Binding Cache and sends subsequent data DIRECTLY to COA, bypassing HA completely: CN → FA → MN.

UNIT 6: GSM — Global System for Mobile Communication

Q17: What is GSM? In which frequency bands does it operate?

✅ **Answer**

GSM (Global System for Mobile Communication) is a second-generation (2G) digital mobile telephony standard developed by ETSI. It replaced analog 1G systems with digital voice, SMS, and basic data. GSM uses TDMA+FDMA for channel access and introduced the SIM card.

Band	Uplink	Downlink	Region
GSM-900	890–915 MHz	935–960 MHz	Europe, Africa, Asia, Australia
GSM-1800 (DCS)	1710–1785 MHz	1805–1880 MHz	High-density cities alongside GSM-900
GSM-850	824–849 MHz	869–894 MHz	North America, Latin America
GSM-1900 (PCS)	1850–1910 MHz	1930–1990 MHz	North America, Caribbean

East Africa primarily uses GSM-900 and GSM-1800 bands.

Q18: Differentiate between Tele Services and Supplementary Services in GSM.

✅ **Answer**

Aspect	Tele Services	Supplementary Services
Definition	Primary end-to-end services that deliver voice/text/data directly between users.	Optional features provided ON TOP of Tele Services to modify the basic experience.
Purpose	Provide the core communication service itself.	Modify, enhance, or control how the core service behaves.
Examples	Voice calls, Emergency calling (free, no SIM needed), SMS (160 chars), MMS, Fax.	Call Forwarding, Call Hold, Call Waiting, Multi-party (up to 6), Caller ID, Call Barring.
Standalone?	YES — work independently.	NO — cannot exist without a Tele Service.
Example in use	You call a friend and speak — Tele Service (voice call).	You put first call on Hold to answer second — Supplementary Service (Call Hold + Call Waiting).

Q19: With the help of GSM architecture, explain the basic parts of the Network Subsystem (NSS).

✅ Answer — 10 marks

The GSM Network Subsystem (NSS), also called the Core Network, is the 'brain' of GSM. It handles call routing, subscriber management, authentication, billing, and external network connections.

GSM Architecture overview: Mobile Station (MS) ↔ Base Station Subsystem (BSS) ↔ Network Subsystem (NSS) ↔ External Networks (PSTN, Internet)

Component	Full Name	Role	Real-World Analogy
HLR	Home Location Register	PRIMARY subscriber database — stores PERMANENT data: phone number (MSISDN), IMSI, subscription type, services, location, and status. Every subscriber has exactly one HLR entry.	The national ID register — every citizen has a permanent record here.
MSC	Mobile Switching Center	Core telephone exchange — routes calls between mobile, landline (PSTN), and other networks. Handles billing (CDRs) and signaling.	A telephone exchange operator — routes every call to the correct destination.
VLR	Visitor Location Register	TEMPORARY database linked to each MSC. When subscriber enters a new area, data is copied from HLR to VLR for fast local processing. Deleted when subscriber leaves.	A hotel guest register — records visitors temporarily; deletes record on checkout.
AUC	Authentication Center	Provides authentication (A3), encryption key generation (A8), and call encryption (A5). Stores secret Ki key — never transmitted over air.	A passport control officer — verifies identity using secure documents.
EIR	Equipment Identity Register	Database of IMEI numbers: White List (allowed), Grey List (suspect), Black List (stolen/unauthorized/defective — blocked).	A stolen vehicle database — police check before allowing a car; network checks before allowing a phone.

Real-World Example: You travel from Dar es Salaam to Mwanza. Your permanent record stays in the DSM HLR. The Mwanza MSC creates a temporary copy in its VLR for fast service. AUC verifies your SIM is genuine. If your phone was stolen and reported, the EIR Black List prevents it connecting — even with a different SIM.

Q20: How does GSM security work? Explain the A3, A5, and A8 algorithms.

✅ **Answer**

Algorithm	Purpose	Input	Output
A3	Authentication	128-bit RAND challenge + 128-bit Ki	32-bit SRES (Signed Response) — verified against network's own SRES
A8	Session Key Generation	128-bit RAND + 128-bit Ki	64-bit session key Kc — used by A5 for encryption
A5	Encryption (stream cipher)	Kc + frame number	Encrypted bitstream. A5/1=strong; A5/2=weak; A5/3=Kasumi (3G)

GSM also uses TMSI (Temporary Mobile Subscriber Identity) instead of IMSI over the air, protecting subscriber identity.

UNIT 7: GPRS — General Packet Radio Service

Q21: Briefly explain four parameters used to define QoS in GPRS.

✅ **Answer — CBE Q3d — 4 marks**

QoS Parameter	Definition	Classes	Example
Service Precedence (Priority)	Importance of data flow — higher priority packets forwarded first during congestion.	High, Normal, Low	Ambulance alert = High; background app update = Low.
Reliability	Maximum acceptable rates of packet loss, duplication, missequencing, and corruption.	Class 1 (strictest) to Class 3 (tolerant)	Bank transaction = Class 1 (zero loss). Weather widget = Class 3.
Delay	Maximum end-to-end transfer time between sending and receiving GPRS interfaces.	Class 1 (<0.5s), Class 2 (<5s), Class 3 (<50s), Best Effort	Vehicle tracking = Class 1. Bulk email = Class 3.
Throughput	Peak (maximum) and mean (average) bit rate guaranteed for the session.	Peak: 8–2048 kbps options	Video streaming = needs guaranteed mean ≥128 kbps for smooth play.

Q22: Explain the GPRS MS three states (mobility management).

✅ **Answer**

State	Description	Location Tracking
IDLE	MS not registered to GPRS. Not reachable by network.	Unknown — no tracking
READY	MS performed GPRS Attach and is active or recently transmitted. READY timer running.	Full cell-level — MS reports every cell change to SGSN
STANDBY	READY timer expired. MS registered but inactive.	Routing Area (RA) level — MS only reports when it moves to a new RA

UNIT 8: UMTS — Universal Mobile Telecommunications System (3G)

8.1 UMTS Overview

UMTS is the 3G successor to GSM, using W-CDMA (Wideband CDMA) radio access. It is compatible with GSM/GPRS core networks (reuses HLR, MSC/VLR, SGSN, GGSN). Base UMTS supports 384 kbps; HSDPA variants reach 3.6+ Mbps.

8.2 UMTS Three Main Components

a) User Equipment (UE):

- Contains Mobile Equipment (ME) and USIM (UMTS Subscriber Identity Module — stores IMSI, authentication algorithms, encryption keys).

b) UTRAN (UMTS Terrestrial Radio Access Network):

- Node B (Base Stations): Converts data between Uu radio interface and Iub interface. Handles channel coding, interleaving, modulation.
- RNC (Radio Network Controller): Controls radio resources for Node Bs, handles handover, connects to MSC/VLR (circuit-switched) and SGSN (packet-switched).

c) Core Network (CN) — shared with GSM/GPRS:

- Contains: HLR, MSC/VLR, Gateway MSC (to PSTN), SGSN, GGSN (to Internet).

Q23: Describe the four UMTS QoS traffic classes.

✅ **Answer**

Class	Priority	Fundamental Characteristic	Example Applications
Conversational	Highest — most delay-sensitive	Strict, low delay. Preserves timing between stream entities. Interactive real-time pattern.	Voice calls, Video telephony, Video games
Streaming	High	Preserves timing between information entities. Less strict than Conversational.	Streaming multimedia, Video on demand
Interactive	Medium	Request-response pattern. Preserves data integrity. Some delay acceptable.	Web browsing, Network games
Background	Lowest — least time-sensitive	Destination not time-sensitive. Data integrity preserved.	Background email download, File transfers

📝 **Exam Tip:** Memorize all 4 UMTS QoS classes in order: Conversational → Streaming → Interactive → Background. 'Conversational' = most strict on delay; 'Background' = least strict.

UNIT 9: 4G LTE — Long Term Evolution

9.1 What is 4G?

4G LTE represents a system architecture REVOLUTION — not just an upgrade. 4G uses a completely new all-IP network design (called EPS — Evolved Packet System) for the purpose of high mobility and high data rates. Key elements:

- Control Plane: Exchanges signaling messages to control the User Equipment (UE).
- User Plane: Carries actual user data — both processes occur on SEPARATE network elements.

9.2 4G Architecture Components

Term	Definition
eNodeB	Evolved Node B — the 4G base station. Works independently (no central controller needed, unlike 3G's RNC). Direct peer connections with other eNodeBs via X2 interface.
MME (Mobility Management Entity)	Controls signaling; handles authentication, authorization, and session management. Does NOT carry user data.
SGW (Serving Gateway)	Routes and forwards user data packets. Handles inter-eNodeB handover to eliminate data gaps during mobility.
PGW (PDN Gateway)	Connected to external networks (Internet). Assigns IP addresses, applies QoS policies.
HSS (Home Subscriber Server)	Replaces HLR — stores all user subscription data in 4G.

9.3 Applications of 4G

- Data-intensive interactive services — online satellite mapping loads instantly.
- Multiple-user video conferencing — subscribers see and talk to more than one person simultaneously.
- Location-based services — real-time weather/traffic, finding nearby businesses while communicating.
- Tele-medicine — medical provider monitors or advises remote patients, streaming video guides.
- HDTV streaming — high-definition TV channels redirected directly to the subscriber's device.

UNIT 10: 5G Networks — Fifth Generation

10.1 Key Specifications

Feature	5G Specification	Compared to 4G
Speed	Up to 20 Gbps	~100x faster than 4G
Latency	As low as 1 millisecond	4G = 30–50ms; 5G is 50x lower
Devices	1 million devices per km²	4G = only ~4,000 per km²
Energy	90% less energy per bit	Far more efficient
Reliability	99.999% uptime	Critical for life-saving systems

10.2 Three 5G Service Types — MOST IMPORTANT for exams

Service	Full Name	Priority	Applications	Real Example
eMBB	Enhanced Mobile Broadband	Maximum SPEED	4K/8K video, AR/VR, cloud gaming	80,000 fans streaming a stadium match in HD simultaneously — 5G eMBB handles all without slowdown.
URLLC	Ultra-Reliable Low-Latency Communication	Minimum DELAY + Maximum RELIABILITY	Self-driving cars, remote surgery, industrial robots	Surgeon in London controls robotic arm performing surgery in Nairobi — any delay over 5ms could cost the patient's life. 5G URLLC makes this safe.
mMTC	Massive Machine-Type Communications	Maximum DEVICE DENSITY	Smart meters, agriculture sensors, smart city	Smart city with 500,000 sensors per km² monitoring traffic, air quality, water — all connected via 5G mMTC.

📝 **Exam Tip:** Memory: eMBB = SPEED (your phone). URLLC = LIFE-CRITICAL (surgery/cars). mMTC = BILLIONS of small IoT devices.

10.3 Key 5G Technologies

Technology	Explanation	Real Example
mmWave	24–100 GHz. Extremely fast but short range (~100-200m). Blocked by walls.	5G mmWave in a stadium — antennas installed every 100m.
Sub-6 GHz	Below 6 GHz. Good range, penetrates walls. Wide area coverage like 4G but with 5G features.	Sub-6 GHz 5G covering rural Tanzania with better speed.
Massive MIMO	Base station uses 64–256+ antennas simultaneously to serve many users.	5G tower in a CBD serving hundreds of users simultaneously.
Beamforming	Focuses signal like a spotlight directly at each user instead of broadcasting in all directions.	Signal always directed at you — not wasted in empty space.
Network Slicing	Divides one physical 5G network into multiple virtual networks, each configured for a specific purpose.	Ambulance slice (fastest/lowest latency) + consumer slice + IoT slice — all isolated.
Edge Computing (MEC)	Processing happens near user at base station — not in distant cloud.	Self-driving car processes collision data in 1ms at local base station vs. 50ms to distant cloud.

UNIT 11: Cloud Computing

11.1 Definition and 5 Characteristics

Cloud Computing is the delivery of computing services — servers, storage, databases, software — over the internet. Instead of buying hardware, you rent computing power from providers (AWS, Google, Microsoft) and pay only for what you use.

Analogy: Electricity — you don't build your own power station. You plug into the national grid and pay per unit used.

Characteristic	Meaning	Real Example
On-Demand Self-Service	Get resources automatically — no human from provider needed.	Startup provisions 10 servers via AWS console in 5 minutes.
Broad Network Access	Access from anywhere — phone, laptop, tablet.	Student in Mwanza accesses Dar es Salaam university library from their phone.
Resource Pooling	Provider serves many customers on shared hardware (each customer's data is isolated).	AWS serves millions of businesses from shared data centers.
Rapid Elasticity	Scale up or down instantly.	Black Friday: Amazon adds thousands of servers at midnight, removes them by morning.
Measured Service	Pay only for what you use.	Company pays for exactly 500GB used this month, 700GB next — no fixed cost.

11.2 Three Cloud Service Models

Model	Full Name	Who Manages What	Best For	Real Example
IaaS	Infrastructure as a Service	You manage: OS, software, apps. Provider manages: hardware, virtualisation, networking.	IT companies wanting full control.	Kenyan startup rents Amazon EC2 virtual server for KES 2,000/month instead of buying for KES 200,000.
PaaS	Platform as a Service	You manage: application code only. Provider manages: everything	Developers.	Developer builds mobile banking app on Heroku — never touches a server.

		underneath.		
SaaS	Software as a Service	You manage: NOTHING. Provider manages: everything.	Everyday users.	Company uses Microsoft 365 — 500 employees access Word/Excel from any device; IT never manages updates.

📝 **Exam Tip:** Exam analogy that ALWAYS earns marks: IaaS = empty apartment (you furnish it). PaaS = furnished apartment (just move in). SaaS = hotel room (everything done for you).

UNIT 12: Internet of Things (IoT)

12.1 Definition

IoT is the network of physical objects — machines, appliances, sensors, vehicles — embedded with sensors and software that connect to the internet and automatically collect and exchange data WITHOUT human involvement.

Formula: IoT = Smart Object + Internet Connection + Automatic Data + Automatic Action.

12.2 How IoT Works — 4 Steps

Step	What Happens	Real Example
1 — Sense/Collect	Sensor detects and measures physical condition.	Soil moisture sensor detects moisture has dropped to 20% — too dry for crops.
2 — Connect/Send	Measured data transmitted wirelessly to server.	Sensor sends reading via LoRaWAN to cloud server 10km away.
3 — Process/Analyze	Cloud/edge computing analyzes the data.	Cloud software checks: moisture too low + hot weather forecast = irrigation needed.
4 — Act/Automate	Automatic decision and action executed.	Irrigation system switches on for 20 minutes. Farmer receives alert on phone.

12.3 Applications

Area	How IoT Helps	Real Example
Healthcare	Continuous remote patient monitoring without clinic visits.	Apple Watch detected a user's atrial fibrillation — alerted user to see cardiologist. Early detection potentially saved their life.
Smart City	Thousands of sensors manage traffic, waste, parking, air quality automatically.	Barcelona IoT bins signal when 80% full → garbage trucks only visit full bins → 40% route reduction, 37% fuel savings.
Agriculture	Sensors, drones, and automated systems for precision farming.	California vineyards with 1,000+ soil sensors per farm → 25% less water used, improved grape quality.
Smart Home	Automated home tasks based on sensors and learning.	Google Nest thermostat learns schedule → starts warming at

		5:45pm, stops heating when you go on holiday.

12.4 IoT Security Challenge

IoT devices are often poorly secured: limited processing (strong encryption difficult), shipped with default passwords users never change, millions of unpatched devices.

Real Example — Mirai Botnet (2016): Hackers found 600,000 IoT devices (cameras, routers) with default passwords. Infected all with malware. Used them to flood Dyn DNS company — took down Twitter, Netflix, Amazon, Reddit for hours. Caused entirely by unsecured IoT devices.

UNIT 13: MANETs — Mobile Ad Hoc Networks

Q24: How does MANET differ from WLAN?

✅ **Answer — CBE Q5a — 4 marks**

Feature	WLAN (Wireless LAN)	MANET (Mobile Ad Hoc Network)
Infrastructure	REQUIRED — access points, routers, wired backbone installed before use.	NONE — devices connect directly without any pre-installed hardware.
Device role	END USERS only — communicate through APs.	BOTH user AND router — each device forwards data for others (multi-hop).
Setup time	Hours to days — IT administrator configures.	Seconds to minutes — automatic self-configuration.
Centralized control	YES — access point controls all transmissions.	NO — fully decentralized.
Failure point	AP failure = entire network down.	Distributed — automatic re-routing around failures.
Best used for	Offices, homes, campuses.	Disaster zones, military, vehicles, areas with no infrastructure.

Q25: Briefly describe THREE applications of Mobile Ad-Hoc Networks.

✅ **Answer — CBE Q5b — 3 marks**

1. Tactical Military Networks:

Military forces in remote terrain need communication without fixed infrastructure that could be targeted or destroyed. Each soldier's radio device connects to nearby devices, forming a multi-hop network. Messages hop soldier-to-soldier to reach the commander. If a device is destroyed, the network automatically re-routes around it.

2. Emergency Services and Disaster Recovery:

Natural disasters (earthquakes, floods) destroy communication infrastructure. Rescue teams arrive with MANET-capable devices and within minutes have a working network across the disaster zone. Example: The 2010 Haiti earthquake destroyed all towers — international rescue teams used MANET radios to coordinate immediately.

3. Vehicle-to-Vehicle (V2V) Communication:

Vehicles form a MANET to share real-time safety information — hazards, accidents, road conditions. A car detects black ice and broadcasts an alert that hops through vehicles behind it via MANET in milliseconds — drivers kilometres back receive the warning before reaching the hazard.

Q26: Explain FOUR constraints facing Mobile Ad-Hoc Networks.

✅ **Answer — CBE Q5c — 8 marks**

1. Infrastructure-less — New Network Design Challenges:

No routers, APs, DHCP servers, or DNS. All networking functions must be distributed across devices. Every device participates in routing, address management, and organization simultaneously. Challenge: protocols must be designed from scratch for a decentralized environment with no central authority.

2. Dynamically Changing Topologies:

Devices move constantly — the network map changes every few seconds. Routes valid now may be broken moments later. Consequences: route breakage (transfers interrupted), network partitions (isolated segments), packet loss. Example: A rescue worker walking into a collapsed building breaks the MANET route through them — all traffic on that path is lost.

3. Physical Layer Limitations:

- Limited wireless range — direct communication only ~100–300m; longer requires multi-hop.
- High packet loss — radio interference, walls, distance cause far higher error rates than wired.
- Broadcast nature — all nearby devices receive all transmissions — interference and privacy risks.

4. Limited Mobile Node Resources:

- Battery drain — acting as a routing hub for 20 other devices drains battery far faster than being just a user.
- Limited CPU — complex routing calculations are difficult on low-power embedded devices.
- Limited memory — routing tables for hundreds of nodes require storage constrained devices may not have.

Q27: For effective bandwidth utilization, which MANET routing protocol is recommended? Justify.

✅ **Answer — CBE Q5d — 5 marks**

Recommendation: REACTIVE (ON-DEMAND) routing protocol — specifically AODV (Ad Hoc On-Demand Distance Vector).

Justification:

The wireless channel in MANET is limited and shared. Routing protocols generating continuous control traffic waste this scarce bandwidth even when no actual data is being sent.

Why PROACTIVE protocols (e.g., DSDV) WASTE bandwidth: Every device continuously broadcasts routing updates to ALL neighbours — whether or not any data is being sent. In a 100-

device network, each device constantly receives 99 routing updates. Channel is busy with routing traffic even during complete silence.

Why AODV is BETTER:

- Routes only discovered WHEN data actually needs to be sent.
- Zero routing messages when no data flows — entire bandwidth preserved for actual data.
- Route Request (RREQ) flood only triggered on demand.
- Once a route is established, data travels directly — no ongoing routing overhead.
- Scales efficiently: 500-node network doesn't flood itself when only 5 nodes communicate.

AODV Operation: Source broadcasts RREQ → Intermediate nodes rebroadcast if no route → Destination sends RREP back along reverse path → Source uses discovered route → If link breaks, RERR sent, route rediscovered.

Conclusion: AODV is best for bandwidth efficiency because it generates routing traffic ONLY when needed, preserving scarce wireless bandwidth entirely for actual data during idle periods.

UNIT 14: CBE Past Paper — Complete Solved Examination

Instructions: FIVE questions. Attempt ALL. Each question = 20 marks.

GREEN boxes = actual CBE exam questions exactly as set. White answers follow immediately below.

QUESTION ONE — 20 MARKS

Q1a [CBE Exam]: Question 1(a) — 2 marks Define the term Mobile Computing.

✅ **Answer — 2 marks**

Mobile Computing is a technology that allows transmission of data, voice, and video via a computer or any other wireless-enabled device without having to be connected to a fixed physical link.

Real example: A doctor in a moving ambulance accessing a patient's medical records on a tablet — moving, wireless, live data. That is mobile computing.

Q1b [CBE Exam]: Question 1(b) — 9 marks Briefly explain any FOUR characteristics and TWO constraints of mobile computing.

✅ **Answer — 9 marks — 4 characteristics × 1.5 marks + 2 constraints × 1.5 marks**

FOUR CHARACTERISTICS:

- Mobility — Users move from one location to another while remaining connected without interruption. Example: Sales executive checking email while commuting by train.
- Portability — Devices are small and lightweight enough to carry everywhere. Example: Field engineer carrying a tablet to a construction site to access drawings remotely.
- Wireless Communication — Relies on WiFi, 3G/4G/5G, Bluetooth — no physical cables. Example: Browsing internet on phone using 4G coverage anywhere with signal.
- Connectivity — Device automatically switches between available networks to maintain connectivity. Example: Smartphone switching from home WiFi to 4G as you leave home — video call continues.

TWO CONSTRAINTS:

- Limited Resources — Less processing power, memory, storage, and battery than desktops. Example: GPS mapping app drains smartphone battery in 3–4 hours vs. unlimited power on desktop.
- Variable Connectivity — Signal strength changes as user moves. Performance varies with location, obstacles, and interference. Example: Clear video call at home breaks up when moving to area with weak signal.

Q1ci [CBE Exam]: Question 1(c)(i) — 5 marks Differentiate between User Mobility and Host Mobility.

✅ **Answer — 5 marks**

Aspect	User Mobility	Host Mobility
Definition	Ability of a PERSON to physically move while continuing to use the same service using the same identity/account.	Ability of a DEVICE (host) — acting as client or server — to move between networks while maintaining reachable network address.
What moves?	The person — device may or may not change.	The device itself moves between networks.
IP handling	IP may change — service follows login credentials.	Must maintain reachable IP using Mobile IP (HA, COA, tunneling).
Complexity	Less complex.	More complex — requires Mobile IP mechanisms.
Example	Student moves Dar to Mwanza, accesses university portal with same username/password.	A server on a vehicle moving between cities that must remain reachable at its permanent IP.

Key distinction: User Mobility = the PERSON continues their service. Host Mobility = the DEVICE maintains its network address and reachability using Mobile IP.

Q1cii [CBE Exam]: Question 1(c)(ii) — 4 marks Differentiate between TDMA and FDMA.

✅ **Answer — 4 marks**

Feature	FDMA	TDMA
How it divides	Divides BANDWIDTH into frequency channels — each user gets own dedicated frequency.	Divides TIME into slots — all users share same frequency, each gets a fixed slot.
Simultaneous use	All transmit at same time on different frequencies.	Only one user per time slot on shared frequency.
Guard mechanism	Guard BANDS between frequencies.	Guard TIMES between slots.
Synchronization	Not required.	Strict timing synchronization required.
Power efficiency	Lower — idle channel wasted.	Higher — unused slots can be reallocated.
Primary use	Early cellular; GSM gross frequency division.	GSM (2G), PDC, IS-136.

GSM combines BOTH: FDMA divides spectrum into 200 kHz channels, then TDMA divides each channel into 8 time slots → 8 users per frequency.

QUESTION TWO — 20 MARKS

Q2a [CBE Exam]: Question 2(a) — 8 marks With illustrations, explain how the Exposed Terminal Problem differs from the Hidden Terminal Problem. Briefly describe the problem in each case.

✅ **Answer — 8 marks**

Both problems arise in wireless MAC when devices share a channel and cannot perfectly detect each other's transmissions.

HIDDEN TERMINAL PROBLEM:

Setup: A ———► B ◄——— C (A and C cannot hear each other; both can reach B)

- Step 1: A begins transmitting to B.
- Step 2: C cannot detect A's transmission — A is out of C's range (hidden).
- Step 3: C falsely senses channel as FREE and also starts transmitting to B.
- Step 4: Both signals arrive at B simultaneously → COLLISION. Both transmissions corrupted.
- Step 5: Neither A nor C knows the collision occurred.

Problem caused: Data loss due to collision. Reduced throughput and reliability. Solution: RTS/CTS — C hears CTS from B and stays quiet even though it could not hear A's RTS.

EXPOSED TERMINAL PROBLEM:

Setup: A ◄——— B ———► C ———► D (D is out of B's range; C can hear B)

- Step 1: B is already transmitting to A.
- Step 2: C wants to transmit to D (D is outside B's range — no interference possible).
- Step 3: C detects B's signal → C incorrectly assumes channel is BUSY and waits.
- Step 4: C's transmission to D would NOT have caused any interference at A.
- Step 5: C waits unnecessarily — wasting a transmission opportunity.

Problem caused: Wasted bandwidth and unnecessary delay. C is 'exposed' to B's signal and mistakenly holds back when it could have transmitted safely.

KEY DIFFERENCE: Hidden terminal → COLLISION occurs because sender cannot detect another's transmission. Exposed terminal → NO collision but sender wrongly stops itself from transmitting.

Q2b [CBE Exam]: Question 2(b) — 6 marks Explain in detail how Delay and Throughput can be used to compare MAC protocols.

✅ **Answer — 6 marks — 3 marks per metric**

1. DELAY (Access Delay):

Definition: The average time a packet spends in the MAC queue — from the moment it arrives at the MAC layer until its transmission is fully complete and acknowledged.

What it includes: Queuing delay, transmission delay, propagation delay, retransmission delay (if collision or error).

How it compares protocols: A protocol causing fewer collisions gives lower average delay. Protocols compared under IDENTICAL traffic conditions — the one with lower mean delay is better for real-time applications.

Important: Delay is sensitive to traffic load — may be low under light load but very high under heavy load. Fair comparison requires identical conditions.

Real example: On a busy airport WiFi, a poor MAC protocol may cause 2-second delay before your request leaves your device — noticeable in video calls. A well-designed MAC keeps this under 50ms.

2. THROUGHPUT:

Definition: The fraction of total channel capacity successfully used for useful data — excluding overhead, collisions, retransmissions, and control packets.

Formula: Throughput $\eta = P / (T \times C)$ where P = packet size, T = elapsed time, C = channel capacity.

How it compares protocols: A protocol with many collisions wastes capacity on retransmissions — throughput drops. Excessive RTS/CTS for tiny packets also wastes capacity. The protocol achieving HIGHER throughput under same conditions is more efficient.

Real example: A 100 Mbps WiFi channel achieves only 60 Mbps actual throughput under a poor MAC protocol — 40% wasted on collisions and retransmissions.

The Trade-off: These metrics often CONFLICT. Reducing delay aggressively (transmit immediately) increases collisions → reduces throughput. Increasing throughput (careful coordination) may increase delay. A well-designed MAC must BALANCE both.

Q2c [CBE Exam]: Question 2(c) — 6 marks How are RTS and CTS used to avoid collision in multiple access?

✅ Answer — 6 marks — refer to detailed answer in Unit 1, Q4 above

RTS (Request to Send) and CTS (Clear to Send) are signalling packets used by IEEE 802.11 MAC to reserve the wireless channel before sending actual data — preventing collisions especially from hidden terminals.

- Step 1: Sender transmits short RTS to receiver [contains: sender address, receiver address, duration].
- Step 2: Receiver replies with CTS — broadcast to ALL nearby devices [also contains duration].
- Step 3: ALL devices hearing CTS set their NAV timer and stay silent for the reserved duration.
- Step 4: Sender transmits the actual data packet — channel reserved, no interference.

- Step 5: Receiver sends ACK. Channel released.

Why it prevents collisions: Device C (hidden from A) hears B's CTS → sets NAV → stays quiet even though it never heard A's RTS. If RTS/CTS collide, only tiny 20-byte control frames are wasted vs. large data frames.

QUESTION THREE — 20 MARKS

Q3a [CBE Exam]: Question 3(a) — 2 marks What is GSM? In which frequency bands does GSM operate?

✅ Answer — 2 marks — see Unit 6 Q17 above for full table

GSM (Global System for Mobile Communication) is a 2nd generation (2G) digital mobile telephony standard by ETSI, using TDMA+FDMA. It replaced analog 1G with digital voice, SMS, and basic data. Frequency bands: GSM-900 (890–960 MHz), GSM-1800 (1710–1880 MHz), GSM-850 (824–894 MHz), GSM-1900 (1850–1990 MHz). East Africa primarily uses GSM-900 and GSM-1800.

Q3b [CBE Exam]: Question 3(b) — 4 marks Differentiate between Tele Services and Supplementary Services in GSM.

✅ Answer — 4 marks — see Unit 6 Q18 above for full table

Tele Services are primary end-to-end communication services (voice calls, SMS, emergency calling, fax). Supplementary Services are optional enhancements ON TOP of Tele Services (call forwarding, hold, waiting, conference, caller ID). Key distinction: Tele Services can exist alone; Supplementary Services CANNOT exist without a Tele Service.

Q3c [CBE Exam]: Question 3(c) — 10 marks With the help of GSM architecture, explain the basic parts of the Network Subsystem (NSS).

✅ Answer — 10 marks — ~2 marks per component — see Unit 6 Q19 for complete answer with table

The NSS (Network Subsystem), also called the Core Network, is the 'brain' of GSM. It handles call routing, subscriber databases, authentication, billing, and external network connections. The five key components are:

- HLR (Home Location Register): PRIMARY subscriber database — permanent data: MSISDN, IMSI, subscription, location, status. Analogy: National ID register.
- MSC (Mobile Switching Center): Core telephone exchange — routes calls, handles billing (CDRs), signaling. Analogy: Telephone exchange operator.
- VLR (Visitor Location Register): TEMPORARY database per MSC — copy of subscriber data from HLR for fast local service. Deleted when subscriber leaves. Analogy: Hotel guest register.
- AUC (Authentication Center): Runs A3 (authentication), A8 (key gen), A5 (encryption). Stores secret Ki — never transmitted over air. Analogy: Passport control officer.

- EIR (Equipment Identity Register): IMEI database — White (allowed), Grey (suspect), Black (stolen/blocked). Analogy: Stolen vehicle database.

Q3d [CBE Exam]: Question 3(d) — 4 marks Briefly explain four parameters used to define QoS in GPRS.

✅ **Answer — 4 marks — 1 mark per parameter — see Unit 7 Q21 above for full table**

The four GPRS QoS parameters are: (1) Service Precedence — priority level (High/Normal/Low) for packet forwarding during congestion. (2) Reliability — maximum acceptable packet loss, duplication, missequencing, and corruption rates (Class 1 strictest to Class 3). (3) Delay — maximum end-to-end transfer time (Class 1 <0.5s to Best Effort). (4) Throughput — guaranteed peak and mean bit rates for the session.

QUESTION FOUR — 20 MARKS

Q4a [CBE Exam]: Question 4(a) — 6 marks What is Care-of Address (COA) in Mobile IP? Briefly explain its two types.

✅ **Answer — 6 marks — see Unit 5 Q15 above for full answer**

A Care-of Address (COA) is a temporary IP address assigned to a Mobile Node when visiting a foreign network. It represents the MN's CURRENT LOCATION. The Home Agent uses the COA to tunnel data to the MN wherever it is. Two types:

- Foreign Agent COA (FA-COA): The FA router's IP address serves as the COA. SHARED — multiple MNs can use same FA-COA simultaneously. Saves IP addresses.
- Co-located COA (CCoA): Unique temporary IP assigned DIRECTLY to MN's own network interface via DHCP. UNIQUE per MN. Works even with no FA in foreign network. Enables route optimization.

Q4b [CBE Exam]: Question 4(b) — 3 marks Briefly describe THREE properties of Care-of Address.

✅ **Answer — 3 marks — 1 mark per property**

- Foreign-link specific: A COA is valid only within the particular foreign network currently being visited. The moment MN moves to a different network, the COA is no longer valid.
- Dynamically changing: COA generally changes every time MN moves to a new foreign network. The HA must always have the most current COA to forward traffic correctly.
- Exit-point of HA tunnel: COA is the termination point of the tunnel created by the Home Agent. HA encapsulates packets (addressed to MN's permanent home IP) inside a new tunnel packet whose destination is the COA. COA appears in the outer (tunnel) packet header.

Q4c [CBE Exam]: Question 4(c) — 8 marks Explain problems of triangular routing and the solution.

✅ **Answer — 8 marks — see Unit 5 Q16 above for complete answer**

Triangular Routing: CN → HA → FA → MN. Data always routes through HA even when CN and MN are geographically nearby.

Problems: (1) Sub-optimal/inefficient routing. (2) Increased latency — extra hops through HA degrade real-time applications. (3) HA bottleneck — all roaming traffic through one point. (4) Firewall problem — foreign network's ingress filter rejects MN's home IP as source. (5) Multicast problem — MN loses group membership. (6) TTL problem — packet TTL may expire on longer triangular path.

Solution — Reverse Tunneling: Creates reverse tunnel MN → FA → HA → CN for outgoing traffic. Fixes firewall (packets appear from HA), multicast (routed through home network), and TTL issues.

Better solution — Route Optimization (RFC 3775): MN sends Binding Update directly to CN. CN stores COA in Binding Cache, sends future data directly: CN → FA → MN. Completely bypasses HA.

Q4d [CBE Exam]: Question 4(d) — 3 marks Differentiate between Mobility and Nomadicity.

✅ **Answer — 3 marks — see Unit 4 Q14 above for full table**

Mobility = the capability to physically move between locations/networks while maintaining connectivity (focus on the ACT of movement). Nomadicity = computing paradigm where the entire environment (apps, data, preferences) travels WITH the user — accessible from any location or device (focus on CONTINUITY of computing experience). Mobility requires Mobile IP; Nomadicity requires cloud storage, VPNs, and device-independence.

QUESTION FIVE — 20 MARKS

Q5a [CBE Exam]: Question 5(a) — 4 marks How does MANET differ from WLAN?

✅ **Answer — 4 marks — see Unit 13 Q24 above for full table**

Key differences: WLAN requires fixed infrastructure (access points) — MANET needs NONE. In WLAN, devices are only end users — in MANET, every device is BOTH user AND router. WLAN has a single point of failure (AP) — MANET is fully distributed. WLAN is best for permanent locations; MANET for temporary/emergency environments without infrastructure.

Q5b [CBE Exam]: Question 5(b) — 3 marks Briefly describe THREE applications of Mobile Ad-Hoc Networks.

✅ **Answer — 3 marks — 1 mark per application — see Unit 13 Q25 above**

- Tactical Military Networks — communication without fixed infrastructure that could be targeted. Soldiers form multi-hop network automatically.
- Emergency Services and Disaster Recovery — immediate communication after disasters destroy towers. Example: Haiti 2010 earthquake — MANET radios for rescue coordination.
- Vehicle-to-Vehicle (V2V) Communication — vehicles share real-time safety information (hazards, accidents) without roadside infrastructure for every message.

Q5c [CBE Exam]: Question 5(c) — 8 marks Explain any FOUR constraints facing Mobile Ad-Hoc Networks.

✅ **Answer — 8 marks — 2 marks per constraint — see Unit 13 Q26 for detailed answers**

- Infrastructure-less nature — all traditional networking functions (routing, addressing) must be distributed. No central authority. Dramatically increases protocol complexity.
- Dynamically changing topologies — devices move constantly, routes break, network partitions occur, packets in transit are lost. Routing protocols must rediscover paths in milliseconds.
- Physical layer limitations — limited range, high packet loss, broadcast interference, privacy risks from shared medium.
- Limited node resources — battery-powered; acting as routing hub drains battery far faster. Limited CPU for complex routing calculations. Limited memory for routing tables.

Q5d [CBE Exam]: Question 5(d) — 5 marks For effective bandwidth utilization, which routing protocol can be used in MANETs? Justify.

✅ **Answer — 5 marks — see Unit 13 Q27 above for complete AODV answer**

Best choice: AODV (Ad Hoc On-Demand Distance Vector) — a REACTIVE (on-demand) routing protocol.

Justification: AODV generates routing traffic ONLY when data needs to be sent. Zero routing messages when no data flows → entire bandwidth preserved for actual data during idle periods. Scales efficiently — a 500-node network doesn't flood itself when only 5 nodes are communicating.

Compare: Proactive protocols (DSDV) continuously broadcast routing updates to all neighbours — channel busy with routing traffic even during silence.

Note: Hybrid ZRP (Zone Routing Protocol) offers a middle ground — proactive within a local zone, reactive outside — suitable for large MANETs needing both local speed and global efficiency.

UNIT 15: MANET Routing Protocols — Proactive, Reactive & Hybrid

15.1 Overview

MANET routing protocols must work WITHOUT any fixed infrastructure. They are categorised into three families based on when and how they build routing tables.

Feature	Proactive (Table-Driven)	Reactive (On-Demand)	Hybrid
How routes are built	Maintained CONTINUOUSLY — routes to ALL destinations kept up-to-date at all times.	Built ONLY when data needs to be sent — routes discovered on demand.	Proactive within a local zone; Reactive outside the zone.
Control overhead	HIGH — constant routing updates broadcast even when no data flows.	LOW — no routing traffic when no data is being sent.	MEDIUM — combines both costs.
Route discovery delay	ZERO — route already known when needed.	Some delay — must discover route before first packet can be sent.	Zero locally; small delay for far destinations.
Bandwidth usage	Wastes bandwidth on routing updates during silence.	Preserves bandwidth — uses it ONLY for actual data plus on-demand discovery.	Balanced.
Best for	Small, slow-moving networks where routes change rarely.	Large networks with sporadic, bursty traffic.	Large networks needing both local speed and global efficiency.
Examples	DSDV (Destination-Sequenced Distance Vector), OLSR	AODV (Ad Hoc On-Demand Distance Vector), DSR	ZRP (Zone Routing Protocol)

Q28: True or False: Reactive routing protocols have higher control overhead than proactive protocols because they update tables constantly.

✅ **Answer — Answer: FALSE □**

This statement is FALSE. It is PROACTIVE (table-driven) protocols that have higher control overhead because they continuously update routing tables to ALL destinations — whether or not any data is being sent. REACTIVE protocols generate routing traffic ONLY when data needs to be sent, giving them LOWER overhead.

Correction: 'Proactive routing protocols have higher control overhead than reactive protocols because they update tables constantly.' ✓

Q29: True or False: Ad-hoc On-demand Distance Vector (AODV) is an example of a PROACTIVE routing protocol.

✅ **Answer — Answer: FALSE ✗**

This statement is FALSE. AODV (Ad Hoc On-Demand Distance Vector) is a REACTIVE (on-demand) routing protocol. The name itself reveals this — 'On-demand' means routes are only discovered WHEN data needs to be sent. AODV does NOT maintain routing tables to all destinations continuously.

Examples of PROACTIVE protocols: DSDV, OLSR. Examples of REACTIVE protocols: AODV, DSR.

Q30: True or False: Zone Routing Protocol (ZRP) uses a proactive approach for nearby nodes and a reactive approach for distant nodes.

✅ **Answer — Answer: TRUE □**

This statement is TRUE. ZRP (Zone Routing Protocol) is a HYBRID protocol. It divides the network into zones: within the local zone, ZRP uses a proactive approach (routes maintained continuously for fast local communication). For nodes OUTSIDE the zone, ZRP uses a reactive approach (routes discovered on demand to save bandwidth). This makes ZRP efficient for large networks needing both local speed and global bandwidth efficiency.

15.2 MANET Routing Types — Unicast, Broadcast, Multicast

Routing Type	Definition	MANET Example
Unicast	ONE source sends to ONE specific destination node.	A soldier sends a command message to a specific team leader.
Broadcast	ONE source sends to ALL nodes in the network.	A commander broadcasts an alert to every soldier's device simultaneously.
Multicast	ONE source sends to a SPECIFIC GROUP of nodes (not all).	A commander sends mission details to only the 5 members of one squad.
Geocast	Sends to all nodes within a GEOGRAPHIC AREA.	Warning message sent to all vehicles within a 500m radius of an accident.

Q31: True or False: Unicast routing in a MANET refers to sending a message from one source to all other nodes in the network.

✅ **Answer — Answer: FALSE ✗**

This statement is FALSE. Unicast routing sends a message from ONE source to ONE specific destination. Sending to ALL other nodes is called BROADCAST (or flooding). Sending to a defined group of nodes is called MULTICAST.

UNIT 16: Cloud Deployment Models — Deep Dive

16.1 What Does a Deployment Model Describe?

A cloud deployment model primarily describes the LOCATION and MANAGEMENT of the cloud's infrastructure — specifically: who owns it, who can access it, and where it physically sits. It does NOT describe programming languages, billing methods, or what is being built — it describes the CONTROL and ACCESS structure.

Deployment Model	Definition	Who Owns/Controls	Who Can Access	Real Example
Private Cloud	Cloud infrastructure used EXCLUSIVELY by one organization. Hosted on-premises or by a dedicated provider.	Single organization — full control.	Only that organization's authorized users.	A hospital running patient data on its own private servers — no external sharing of hardware.
Public Cloud	Cloud infrastructure owned and operated by a third-party provider, shared across many customers.	Provider (AWS, Azure, Google).	Anyone who subscribes/pays.	A startup using AWS — shares physical servers with thousands of other businesses (data is isolated, hardware is shared).
Hybrid Cloud	Combination of private and public clouds that are integrated — data and applications can move between them.	Both — organization controls private part; provider controls public part.	Depends on which component is accessed.	A company uses its own private servers for sensitive data normally, but BURSTS into AWS during peak traffic (e.g., Black Friday sale).
Community Cloud	Shared infrastructure for a SPECIFIC COMMUNITY of organizations with common concerns (e.g., security requirements, compliance).	Shared by community members or a third party.	Only the specific community members.	Multiple government agencies sharing a cloud that meets specific security regulations.

Q32: An organization uses its own on-premises infrastructure for daily operations but bursts into a public cloud during peak traffic. This is an example of: A. Private Cloud B. Public Cloud C. Hybrid Cloud D. Community Cloud

✅ Answer — Answer: C — Hybrid Cloud

This is a Hybrid Cloud because the organization uses BOTH its own private infrastructure (on-premises, for normal operations) AND a public cloud (for peak traffic bursting). The key indicator is 'bursts into a public cloud' — the private and public components work TOGETHER. This is called cloud bursting, a common Hybrid Cloud pattern.

Q33: What is a primary disadvantage of a Private Cloud compared to a Public Cloud? A. Lower security B. Limited customization C. Higher upfront capital expenditure D. Shared hardware resources

✅ Answer — Answer: C — Higher upfront capital expenditure

The primary disadvantage of a Private Cloud is the HIGHER UPFRONT COST. Organizations must purchase, install, and maintain their own hardware and software — large capital expenditure before any benefit. Public Cloud requires no upfront investment — you pay only for what you use (operational expenditure).

Why the other options are wrong: Private Cloud has BETTER security (not lower) than Public. Private Cloud has MORE customization (not limited). Private Cloud does NOT share hardware — that's a characteristic of Public Cloud.

UNIT 17: Mobile Edge Computing (MEC) & Serverless Computing

17.1 Mobile Edge Computing (MEC)

Mobile Edge Computing moves computational power FROM distant centralized cloud data centers TO the EDGE of the network — physically close to users at or near the base station / access point. This dramatically reduces the distance data must travel, cutting latency.

Term	Definition
MEC / Edge Computing	Processing happens at the EDGE of the network (near the user) rather than in a distant central cloud.
Inferencing	The process of running a trained AI model on NEW data to produce predictions or decisions. In MEC, inferencing happens at the edge node/server — NOT in the central cloud.
Edge Node/Server	The computing hardware deployed at or near the base station that performs edge processing.
CUPS	Control and User Plane Separation — 5G principle that allows control logic and data traffic to be managed on independent, scalable infrastructure.
Round-trip latency	The time for data to travel from device to server and back. Edge computing minimizes this by placing servers near the user.

Q34: Which 5G network feature is MOST CRITICAL for enabling the ultra-low latency required by MEC applications? A. Massive MIMO B. Network Slicing C. Control and User Plane Separation (CUPS) D. Beamforming

✅ **Answer — Answer: C — Control and User Plane Separation (CUPS)**

CUPS (Control and User Plane Separation) is the most critical 5G feature for MEC latency because it allows the USER PLANE (actual data traffic) to be placed at the network edge — physically close to the user — while the CONTROL PLANE remains centralized. This means user data travels a VERY SHORT path to the nearby edge server, achieving ultra-low latency (1ms).

Without CUPS: All data must travel through the central core network — even if the processing could happen at a nearby edge server. With CUPS: Data takes the shortest possible path to the edge node.

Q35: In a Mobile Edge Computing environment, where does 'Inferencing' for an AI model typically occur? A. Centralized Cloud B. Edge Node/Server C. User Equipment (UE) only D. Core Network

✅ **Answer — Answer: B — Edge Node/Server**

In MEC, AI inferencing (running a trained model to make real-time predictions on new data) occurs at the EDGE NODE/SERVER — the computing hardware deployed at or near the base station. This avoids the round-trip delay to a distant centralized cloud.

Real Example: A self-driving car needs to detect a pedestrian in 1ms. If inferencing happened in a centralized cloud (50ms round trip), the car would have already moved. With MEC, the trained detection model runs at the nearby edge server — result returned in 1ms — car stops safely.

Q36: What is the primary driver for moving computational power to the edge in Mobile Edge Computing? A. Increase storage of mobile devices B. Reduce network congestion and round-trip latency C. Eliminate need for cellular base stations D. Provide free internet

✅ **Answer — Answer: B — To reduce network congestion and round-trip latency**

The PRIMARY driver for MEC is reducing LATENCY and CONGESTION. By processing data near the user instead of routing it to a distant cloud: (1) Round-trip time drops from 50–100ms (cloud) to ~1ms (edge). (2) Network congestion is reduced — only processed results (not raw data) travel the wider network. This is essential for autonomous vehicles, remote surgery, and industrial robots — all requiring sub-10ms response times.

17.2 Serverless Computing / Function as a Service (FaaS)

Serverless Computing is a cloud execution model where the cloud provider dynamically manages the allocation of server resources. Developers write small functions (pieces of code) that run ONLY when triggered — the provider handles all server management. Despite the name, servers still exist — the developer just never sees or manages them.

Concept	Description	Real Example
Serverless / FaaS	Code runs in stateless containers triggered by events. Developer writes only the function — provider handles everything else.	An AWS Lambda function that automatically resizes an uploaded image — runs only when a photo is uploaded, charges only for execution time.
Key benefit	No server management. Pay ONLY for execution time (milliseconds), not idle servers.	A startup pays KES 0 when their app has no users, and only tiny amounts per function call when active.
Difference from PaaS	PaaS gives you a platform to run apps continuously. FaaS runs code ONLY on-demand per event.	PaaS: Your app server is always running. FaaS: Code runs only when someone triggers it.

Q37: Which term is often used interchangeably with Serverless computing to describe its execution model? A. IaaS B. PaaS C. Function as a Service D.

SaaS

✅ Answer — Answer: C — Function as a Service (FaaS)

Serverless computing is commonly called Function as a Service (FaaS) because the execution model is based on individual FUNCTIONS (small pieces of code) that run on-demand when triggered by an event. Examples: AWS Lambda, Google Cloud Functions, Azure Functions. The 'serverless' name describes the developer's perspective — they do not see or manage servers, even though servers run their code.

UNIT 18: Artificial Intelligence Applications in Mobile Computing

18.1 Key AI Technologies and Their Applications

AI Technology	Definition	Mobile Computing Application	Real Example
Natural Language Processing (NLP)	AI that enables computers to understand, interpret, and generate human language (text and speech).	Chatbots, voice assistants, automatic translation, sentiment analysis.	A Chatbot on a banking app that understands 'What is my balance?' and responds with account details. Google Translate on mobile.
Computer Vision	AI that enables computers to interpret and understand visual information from images and video.	Facial recognition, object detection, medical imaging, augmented reality.	Face unlock on smartphones. A mobile app scanning a document and extracting text (OCR). Self-driving car cameras.
Robotics	AI combined with mechanical systems to automate physical tasks.	Industrial automation, drone delivery, surgical robots.	Amazon warehouse robots automatically sorting and moving packages. Drone delivery of medicines to remote areas.
Expert Systems	AI that uses a knowledge base and inference rules to make decisions like a human expert in a specific domain.	Medical diagnosis apps, fault diagnosis, financial advisory tools.	A mobile medical app that asks about symptoms and recommends treatment based on a medical knowledge database.
Machine Learning	AI that learns from data to improve predictions without being explicitly programmed.	Recommendation systems, fraud detection, predictive text.	Spotify recommending songs based on your listening history. Auto-complete on your keyboard.

Q38: A Chatbot is an application of which AI technology? A. Robotics B. Natural Language Processing C. Computer Vision D. Expert Systems

✅ **Answer — Answer: B — Natural Language Processing (NLP)**

A Chatbot is powered by Natural Language Processing (NLP) — the branch of AI that allows computers to understand, interpret, and respond to human language in a natural, conversational way. The chatbot processes the USER'S TEXT or SPEECH input (NLP task) and generates an appropriate text/speech response.

Why the others are wrong: Robotics deals with physical mechanical systems. Computer Vision processes images/video (not text/speech). Expert Systems use rule-based knowledge bases — while some chatbots include expert-system elements, the PRIMARY technology enabling a chatbot to understand language is NLP.

UNIT 19: IoT Architecture — The Three Standard Layers

19.1 Standard IoT Three-Layer Architecture

The standard IoT architecture divides functions into three distinct layers, each with a specific role in the data flow from physical sensing to intelligent action.

Layer	Also Called	Primary Responsibility	Components	Real Example
Perception Layer	Sensing Layer / Device Layer	SENSING and GATHERING physical world data. The 'eyes and ears' of IoT.	Sensors (temperature, moisture, motion, RFID, GPS), actuators, cameras, smart meters.	Soil moisture sensor detecting water level in a farm field. RFID tag on a product in a warehouse.
Network Layer	Transport Layer / Connectivity Layer	TRANSMITTING data collected by the Perception Layer to processing systems. The 'nervous system'.	WiFi, 4G/5G, LoRaWAN, Bluetooth, ZigBee, NB-IoT, the internet backbone.	Sending the soil moisture reading via LoRaWAN to a cloud server 10km away.
Application Layer	Service Layer	PROCESSING data and delivering INTELLIGENT SERVICES to end users. The 'brain'.	Cloud platforms, AI/ML analytics, dashboards, automation rules, user apps.	Cloud software analyzing moisture data → deciding irrigation is needed → automatically switching on irrigation system → sending alert to farmer's phone.

Q39: In the standard IoT architecture, which layer is responsible for sensing and gathering data? A. Application Layer B. Network Layer C. Perception Layer D. Support Layer

✅ Answer — Answer: C — Perception Layer

The PERCEPTION LAYER (also called the Sensing Layer or Device Layer) is responsible for sensing and gathering data from the physical world. This is the lowest layer — it contains all physical IoT devices: sensors, actuators, cameras, RFID readers, GPS trackers, and smart meters. These devices convert physical conditions (temperature, motion, moisture, location) into digital data.

Memory tip: Perception = what you PERCEIVE (see, feel, detect) from the physical world.

Q40: Which of the following is a significant security risk for IoT? A. Devices having too many passwords B. Hardcoded default passwords and lack of encryption C. Devices being too fast to track D. Using too much recycled plastic

✅ Answer — Answer: B — Hardcoded default passwords and lack of encryption

The most significant IoT security risk is HARDCODED DEFAULT PASSWORDS combined with LACK OF ENCRYPTION. Most IoT devices ship with simple default credentials (like 'admin/admin' or 'password/password') that users never change. Without encryption, data transmitted by the device is visible in plain text. These two weaknesses together create millions of easily exploitable entry points.

Real Evidence — Mirai Botnet (2016): Hackers scanned for IoT devices with default passwords, found 600,000 unsecured cameras and routers, infected them all, and used them to launch the largest DDoS attack in history — taking down Twitter, Netflix, and Amazon.

UNIT 20: 5G Advanced Architecture — CUPS, UPF, AMF

20.1 5G Core Network Functions

The 5G core uses a Service-Based Architecture (SBA) with separate, independent functions. Understanding WHICH function does WHAT is critical for exam True/False questions.

Term	Definition
AMF (Access and Mobility Management Function)	Handles SIGNALING, authentication, authorization, and mobility management. The 'front door' for UE connection. Does NOT carry user data.
UPF (User Plane Function)	Carries ACTUAL USER DATA (packets) between UE and external networks (Internet). Acts as the data forwarding point. Does NOT do signaling or authentication.
SMF (Session Management Function)	Manages user SESSIONS — assigns IP addresses, selects UPF, applies QoS policies.
AUSF (Authentication Server Function)	Allows AMF to AUTHENTICATE the UE. Works with UDM to verify identity.
UDM (Unified Data Management)	Stores subscriber DATA (like the HLR/HSS in older generations).
PCF (Policy Control Function)	Applies POLICY decisions — QoS rules, charging policies.
CUPS (Control and User Plane Separation)	Core 5G PRINCIPLE separating control plane (signaling/management) from user plane (data traffic) so each can scale independently.

Function	Handles Signaling?	Handles Data Traffic?	Analogy
AMF	YES — authentication, registration, mobility	NO	Airport immigration officer — checks your identity and lets you in.
UPF	NO	YES — forwards all user data packets	The highway — carries all the actual traffic.
SMF	YES — session management	NO	The booking clerk — sets up your session and configures the highway.

Q41: True or False: The User Plane Function (UPF) in 5G handles the signaling and authentication of users.

✅ **Answer — Answer: FALSE ✗**

This statement is FALSE. The UPF (User Plane Function) handles ACTUAL USER DATA TRAFFIC (packets between UE and the internet). It does NOT handle signaling or authentication.

The correct 5G function for signaling and authentication is the AMF (Access and Mobility Management Function). The AUSF (Authentication Server Function) specifically handles authentication. Remember: 'User Plane' = data, NOT signaling.

Q42: True or False: The Mobility Management Entity (MME) is the primary component in the 4G core that handles the actual user data traffic.

✅ **Answer — Answer: FALSE ✗**

This statement is FALSE. In 4G LTE, the MME (Mobility Management Entity) handles SIGNALING — authentication, authorization, and session management control. It does NOT carry user data.

User data traffic in 4G is handled by the SGW (Serving Gateway) and PGW (PDN Gateway). The SGW routes and forwards user data packets between eNodeBs; the PGW connects to external networks (Internet) and assigns IP addresses.

Memory: MME = Management (signaling). SGW/PGW = the data highways.

Q43: True or False: Control and User Plane Separation (CUPS) is a core 5G principle that allows the control logic to be scaled independently from the data traffic.

✅ **Answer — Answer: TRUE ✓**

This statement is TRUE. CUPS is a fundamental 5G architectural principle that separates the CONTROL PLANE (signaling, session management, authentication) from the USER PLANE (actual data forwarding). This separation means: the control plane can scale for millions of signaling operations without needing more data capacity; the user plane can be deployed at the network EDGE for low latency independently of where the control functions run. CUPS is what makes MEC (Mobile Edge Computing) possible — user plane functions can be placed near the user while control remains centralized.

20.2 5G Technology True/False Clarifications

Q44: True or False: Massive MIMO involves using a very small number of antennas to save space on cell towers.

✅ **Answer — Answer: FALSE ✗**

This statement is FALSE. Massive MIMO (Multiple Input Multiple Output) uses a VERY LARGE number of antennas — typically 64 to 256+ antennas at a single base station. The word 'Massive' refers to the massive NUMBER of antennas. This allows the base station to serve many users simultaneously using spatial multiplexing and beamforming, dramatically increasing spectral efficiency. The antennas are dense but compact — integrated into panels on the tower.

Q45: True or False: Beamforming is a technique that broadcasts a signal in all directions equally to ensure maximum coverage.

✅ **Answer — Answer: FALSE ✗**

This statement is FALSE. Beamforming does the OPPOSITE — instead of broadcasting in all directions equally (omnidirectional), beamforming FOCUSES the signal like a SPOTLIGHT directly at a specific user or device. By controlling the phase and amplitude of signals from multiple antennas, beamforming concentrates signal energy in the direction of the intended receiver.

Benefits: Stronger signal at the receiver (better quality), less interference to other users, more efficient use of transmission energy.

Analogy: Traditional antenna = a light bulb illuminating a room equally in all directions. Beamforming = a spotlight that follows you specifically wherever you move.

Q46: True or False: Network Slicing allows a single physical 5G network to be partitioned into multiple virtual networks with different performance characteristics.

✅ **Answer — Answer: TRUE ✓**

This statement is TRUE. Network Slicing is a core 5G capability that creates multiple independent virtual networks (slices) on top of one shared physical infrastructure. Each slice has its own independently configured: bandwidth, latency target, reliability level, and QoS policies. Slices are completely isolated from each other — congestion in one slice does not affect others.

Example: One 5G network sliced into: Slice 1 = ambulance/emergency (guaranteed lowest latency, cannot be congested). Slice 2 = consumer streaming (high bandwidth). Slice 3 = IoT sensors (low power, massive connections). All three run on the SAME physical towers and cables.

UNIT 21: Test Paper 2 — Full Solved (20th February 2026)

ITU073M — MOBILE COMPUTING | TEST 2 | DATE: 20th February 2026 | TIME: 30 minutes

Section 1: Multiple Choice — Complete Answers

Q	Question Summary	Correct Answer	Why
i	What does a deployment model primarily describe in cloud computing?	B. The location and management of the cloud's infrastructure	A deployment model describes WHERE and WHO manages the infrastructure — not what is built, how it's programmed, or how it's billed.
ii	Organization uses on-premises normally but bursts into public cloud during peak — this is:	C. Hybrid Cloud	Uses BOTH private (on-premises) AND public cloud together = Hybrid. The 'bursting' pattern is classic Hybrid Cloud.
iii	Primary disadvantage of Private Cloud vs. Public Cloud?	C. Higher upfront capital expenditure	Private Cloud requires buying/installing own hardware — large capital cost. Public Cloud = pay-as-you-go, zero upfront.
iv	Which 5G feature is most critical for ultra-low latency in MEC applications?	C. Control and User Plane Separation (CUPS)	CUPS allows the User Plane to be placed at the edge near the user — minimizing data travel distance and latency. This is what makes MEC possible.
v	In MEC, where does AI inferencing typically occur?	B. Edge Node/Server	Inferencing runs at the edge node close to the user for sub-millisecond response — NOT in distant centralized cloud.
vi	Primary driver for moving compute power to the edge in MEC?	B. Reduce network congestion and round-trip latency	The core reason is latency reduction and congestion relief — not storage, not eliminating base stations.
vii	Term used interchangeably with Serverless computing?	C. Function as a Service (FaaS)	Serverless = FaaS — code runs as individual functions triggered by

			events. AWS Lambda is the most common example.
viii	In standard IoT architecture, which layer senses and gathers data?	C. Perception Layer	Perception Layer = sensors, cameras, RFID — the physical sensing layer at the bottom of IoT architecture.
ix	Significant security risk for IoT?	B. Hardcoded default passwords and lack of encryption	Default passwords + no encryption = millions of easily hacked devices. This caused the Mirai Botnet (2016) attack.
x	A Chatbot is an application of which AI technology?	B. Natural Language Processing (NLP)	Chatbots understand and generate HUMAN LANGUAGE — the core task of NLP.

Section 2: True/False — Complete Answers with Corrections

Statement #	Statement	Answer	Explanation & Correction if False
i	Reactive routing protocols have higher control overhead than proactive protocols because they update tables constantly.	F — FALSE	PROACTIVE protocols have higher overhead (constant updates). REACTIVE protocols generate routing traffic only on demand.
ii	Ad-hoc On-demand Distance Vector (AODV) is an example of a proactive routing protocol.	F — FALSE	AODV is REACTIVE (on-demand). The name 'On-demand' makes this explicit. Proactive examples: DSDV, OLSR.
iii	Zone Routing Protocol (ZRP) uses a proactive approach for nearby nodes and a reactive approach for distant nodes.	T — TRUE	ZRP is a HYBRID protocol — proactive within local zone, reactive outside. This is its defining characteristic.
iv	Unicast routing in a MANET refers to sending a message from one source to all other nodes in the network.	F — FALSE	Unicast = one source to ONE specific destination. Sending to ALL nodes = Broadcast. To a group = Multicast.

v	Network Slicing allows a single physical 5G network to be partitioned into multiple virtual networks.	T — TRUE	Correct — Network Slicing creates multiple isolated virtual networks on one physical 5G infrastructure with different QoS.
vi	Massive MIMO involves using a very small number of antennas to save space on cell towers.	F — FALSE	Massive MIMO uses 64 –256+ antennas. 'Massive' = massively large number of antennas, NOT small.
vii	Beamforming broadcasts a signal in all directions equally for maximum coverage.	F — FALSE	Beamforming FOCUSES signal like a SPOTLIGHT at a specific user — the opposite of equal omnidirectional broadcasting.
viii	The User Plane Function (UPF) in 5G handles signaling and authentication of users.	F — FALSE	UPF handles USER DATA TRAFFIC. Signaling and authentication are handled by AMF (and AUSF for authentication).
ix	The Mobility Management Entity (MME) is the primary 4G core component handling actual user data traffic.	F — FALSE	MME handles SIGNALING (authentication, session control). User data is handled by SGW (routing) and PGW (gateway to internet).
x	Control and User Plane Separation (CUPS) is a core 5G principle allowing control logic to scale independently from data traffic.	T — TRUE	Correct — CUPS separates control and user planes enabling independent scaling and edge deployment of user plane functions.

📝 **Exam Tip:** TRUE/FALSE EXAM STRATEGY: Read carefully for absolute words like 'all', 'only', 'never', 'always', 'primary', 'small', 'large'. These are often the clue that a statement is FALSE. Always write a one-sentence correction for any FALSE answer to earn full marks.

EXAM STRATEGY — The 4-Step WINNING Formula

Question Word	What It Means	Steps Required	Expected Mark Percentage
Define / Fasiri	Give the meaning of the term.	Step 1 only — be direct, no padding.	~40% if only definition is given
Explain / Eleza	Describe how it works.	Steps 1 + 2 + 3 (example essential).	~80%+ with define + explain + example
Discuss / Jadili	Talk about it in depth.	All 4 steps — earns most marks.	Full marks possible
Compare / Linganisha	Show differences and similarities.	Steps 1 + 2 for EACH item — always use a TABLE.	Always use a comparison table
Justify / Thibitisha	Give reasons why something is used/chosen.	Steps 2 + 3 + 4 — show evidence.	Recommendation + clear justification
Give an example	Show a real-world use.	Step 3 — be specific, name the real thing.	Earns bonus marks on all question types

📝 **Exam Tip:** KEY RULE: When you see 'Explain', 'Discuss', or 'Justify' — ALWAYS include a real-world example. Students who only define get ~40%. Students who define + explain + example get 80%+.

Quick Reference — Key Formulas

Formula	Meaning
$\eta = P / (T \times C)$	Throughput = Packet size / (Time elapsed × Channel capacity)
BER wireless = 10^-3	Wireless Bit Error Rate (much worse than wired 10^-6)
GSM channel	200 kHz (FDMA) × 8 time slots (TDMA) = 8 users per frequency
GPRS max speed	171.2 kbps theoretical; 5–40 kbps realistic
UMTS base speed	384 kbps (HSDPA: 3.6+ Mbps)
4G LTE	Up to ~150 Mbps typical
5G peak speed	Up to 20 Gbps
5G latency	As low as 1 ms (vs. 4G: 30–50ms)
5G device density	1 million devices per km^2 (vs. 4G: ~4,000)

S-MAC active period	Fixed at 115ms
GSM SMS max	160 alphanumeric characters

Good luck in your examination!

Remember: Define → Explain → Real Example → Conclude. Always use tables for comparisons.